Answer **all** the questions in the spaces provided.

1 An elastic rope is attached to a 70.0 kg man on one end and to a bridge on the other end. The rope has a natural length of 25.0 m and an elastic constant of 120 N m^{-1}.

The man steps off the bridge and falls vertically downwards from rest.

Assume that air resistance acting on the man is negligible.

(a) Explain why the person has maximum speed when the tension in the elastic rope is equal to his weight.

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.....

.....

..... [2]

(b) Determine the maximum speed of the man after he steps off the bridge.

maximum speed = ... $m\ s^{-1}$ [3]

(c) Calculate the extension of the elastic rope when the man is at the lowest point of his motion.

extension = ... m [2]

[Total: 7]